AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX

ANNUAL EXAMINATIONS (THEORY) 2025

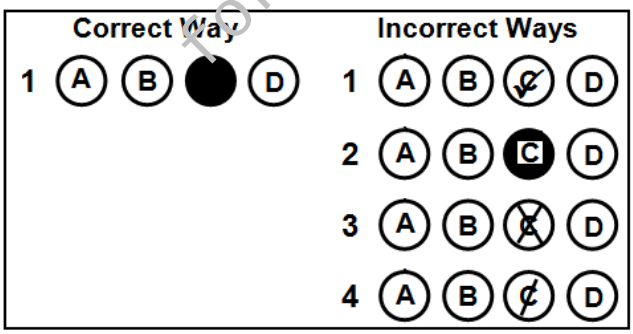
General Mathematics Paper I

Time: 1 hour 20 minutes Marks: 45

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 45 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. A formulae list is provided on page 2. You may refer to it during the paper, if you wish.
8. You may use a simple calculator if you wish.

List of Formulae

Note:

All symbols used in the formulae have their usual meaning.

Business Mathematics

$$\text{Loss }\% = \left(\frac{\text{Loss}}{CP} \times 100\right) \qquad \text{Profit }\% = \left(\frac{\text{Profit}}{CP} \times 100\right) \qquad SP = CP \times \left(\frac{100 + \text{profit}\,\%}{100}\right)$$

$$SP = CP \times \left(\frac{100 - \text{loss}\%}{100}\right) \qquad \text{Discount }\% = \frac{\text{Discount}}{MP} \times 100$$

Sets and Functions

$$(A \cup B)^c = A^c \cap B^c \qquad (A \cap B)^c = A^c \cup B^c$$

Exponents and Logarithms

$$a^m \times a^n = a^{m+n} \qquad a^m \div a^n = a^{m-n} \qquad \frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m$$

$$\left(a^m\right)^n = a^{mn} \qquad (a \times b)^m = a^m \times b^m \qquad a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

$$\log_a (m)^n = n \log_a m \qquad \log_a (m \times n) = \log_a m + \log_a n \qquad \log_a \left(\frac{m}{n}\right) = \log_a m - \log_a n$$

$$\log_a n = \log_b n \times \log_a b \qquad \log_a b = n \Leftrightarrow a^n = b$$

Algebraic Formulae and Applications/ Factorisation

$$(a-b)^2 = a^2 - 2ab + b^2 \qquad (a+b)^2 = a^2 + 2ab + b^2$$

$$a^2 - b^2 = (a+b)(a-b) \qquad 2\left(a^2 + b^2\right) = (a+b)^2 + (a-b)^2$$

$$(a+b)^2 - (a-b)^2 = 4ab \qquad (a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

Linear Graphs

$$1 \text{ mile} = \frac{8}{5}\text{km} \qquad 1 \text{ Hectare} = 2.471 \text{ Acres} \qquad {}^\circ F = \left(\frac{9}{5} \times {}^\circ\text{C}\right) + 32$$

Matrices and Determinants

$$A^{-1} = \frac{1}{|A|} AdjA$$

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1. On converting $\frac{2}{3}$ % of $\frac{3}{2}$ into fraction, we get

   A. $\frac{4}{900}$.
   B. $\frac{1}{100}$.
   C. $\frac{9}{400}$.
   D. $\frac{6}{100}$.

2. If the number of white cars in a showroom is 30 and the ratio of black, white and grey cars is 1:3:4, then the number of grey cars will be

   A. 10
   B. 30
   C. 40
   D. 80

3. Kabir is 15% taller than Umar. If Umar is 150 cm tall, then the height of Kabir will be

   A. 127.5 cm.
   B. 135.0 cm.
   C. 165.0 cm.
   D. 172.5 cm.

4. A person's weight was initially 65 kg. He followed a diet plan and now his current weight's ratio is 12:13 of his previous weight.

   Based on the given information, his current weight is

   A. 60 kg.
   B. 53 kg.
   C. 52 kg.
   D. 40 kg.

5. The magnitude of force of attraction between two bodies decreases with the square of the distance between them. This statement can be translated into mathematical form as

   A. $g \propto r^2$
   B. $g \propto r$
   C. $g \propto \frac{1}{r}$
   D. $g \propto \frac{1}{r^2}$

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6. In the year 2023, Kaif paid an *Ushr* of Rs 100,000 on his agricultural income. His agricultural income for the year 2023 was

   (**Note:** Rate of *Ushr* is 10% of earning.)

   A. Rs 10,000,000
   B. Rs 1,000,000
   C. Rs 100,000
   D. Rs 10,000

7. Salima's annual savings for the year 2023 was Rs 450,000. If the rate of *zakat* is 2.5% per annum, then for the year 2024, the amount left after paying *zakat* will be

   A. Rs 337,500
   B. Rs 425,000
   C. Rs 438,750
   D. Rs 461,250

8. Jamil has two daughters and a son. If he leaves a property worth Rs 1,200,000, then his son's share would be

   (**Note**: The share of a son is two times that of a daughter.)

   A. Rs 300,000
   B. Rs 400,000
   C. Rs 600,000
   D. Rs 900,000

9. Salman buys 12 water bottles from a whole sale dealer for Rs 9,000. If he sells each bottle for Rs 900, then he would make a profit of

   A. Rs 150
   B. Rs 900
   C. Rs 1,800
   D. Rs 10,800

10. Afifa buys a property for Rs 10 million. If she sells it for Rs 11 million, then the percentage profit is

    A. 1%
    B. 10 %
    C. 11 %
    D. 21 %

11. Sajid bought two trousers and two shirts. The tag price of each trouser and shirt is Rs 1,000 and Rs 500 respectively. If the discount offered on the tag price of the trouser and the shirt is 5% and 10% respectively, then the discounted price of two trousers will be

    A. Rs 100
    B. Rs 950
    C. Rs 1,800
    D. Rs 1,900

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12. Meraj bought a shirt for Rs 1,500 after a discount of 20%. The marked/ tag price of the shirt was

    A. Rs 1,875
    B. Rs 1,800
    C. Rs 1,200
    D. Rs 300

13. In a business, Irfan, Sharif and Imran invested Rs 500,000, Rs 300,000 and Rs 200,000 respectively. The profit is to be distributed in the ratio of their individual investments.

    If the business earned a profit of Rs 800,000 in a year, then the share of Sharif will be

    A. Rs 1,000,000
    B. Rs 400,000
    C. Rs 240,000
    D. Rs 160,000

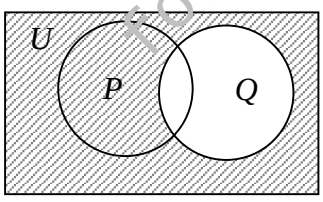
14. If $M = \{1, 2, 3, 4, 5\}$ and $N = \{1, 5\}$, then $M \cap N$ will be

    A. {1, 2, 3, 4, 5}.
    B. {2, 3, 4}.
    C. {1, 3, 5}.
    D. {1, 5}.

15. If A-B = {a, e, i, o, u}, then sets A and B respectively will be

    A. {a, e, i, o, u} and {a, e, i, o, u}.
    B. {a, e, i, o, u} and { }.
    C. { } and {a, e, i, o, u}.
    D. {a, e, i} and {o, u}.

16. The shaded region in the given Venn diagram represents



    A. P'.
    B. Q'.
    C. $P - Q$.
    D. $Q - P$.

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17. If the Cartesian product of two sets is $A \times B = \{(d, 1), (e, 1), (f, 1)\}$, then element(s) of the set B will be

A. $\{1, d, e, f\}$.
B. $\{d, e, f\}$.
C. $\{1\}$.
D. $\{\ \}$.

18. The range of the binary relation $\{(1, 0), (0, 2), (2, 0)\}$ will be

A. $\{0\}$
B. $\{2\}$
C. $\{0, 2\}$
D. $\{0, 2, 0\}$

19. The simplified form of the expression $\left(\frac{1}{x^2}\right)^{-3}$ will be

A. $\frac{1}{x^5}$
B. $\frac{1}{x^6}$
C. x^5
D. x^6

20. In the expression $\frac{1}{\sqrt[3]{2x^3}}$, the radicand is

A. $\frac{1}{3}$
B. 3
C. $2x$
D. $2x^3$

21. On converting the expression $\sqrt[\frac{1}{4}]{x^4}$, the exponential form will be

A. $x^{\frac{1}{16}}$
B. $x^{\frac{1}{4}}$
C. x
D. x^{16}

22. The common notation of the number $2.3 \times \frac{1}{10^3}$ will be

A. 230
B. 2300
C. 0.0230
D. 0.00230

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23. The logarithmic form of the number $\left(\frac{1}{3}\right)^{-2} = 9$ will be

A. $-\log_2 9 = \frac{1}{3}$

B. $-\log_9 2 = \frac{1}{3}$

C. $\log_{\frac{1}{3}} 9 = -2$

D. $\log_9 \frac{1}{3} = -2$

24. In expanded form, $\log_2\left(\frac{1}{8}\right)$ will be equal to

A. -3

B. -1

C. 1

D. 3

25. In multiple logarithms, the expression $\log\left(\frac{x}{y}\right)^4$ can be written as

A. $4\log x - 4\log y.$

B. $4\log x + 4\log y.$

C. $4\log x - \log y.$

D. $4\log(x-y).$

26. Which of the given algebraic expression(s) is/ are rational expression?

I. $x - \sqrt{x}$

II. $x^2 - \frac{1}{x^2}$

III. $x^2 + 2x + 1$

A. I only

B. II only

C. I and III

D. II and III

27. In the simplest form, the expression $\frac{xy}{x-y} \times \frac{(x-y)^2}{x^2}$, can be expressed as

A. $\frac{y(x-y)}{x}.$

B. $\frac{y}{x(x-y)}.$

C. $\frac{y}{x-y}.$

D. $\frac{x-y}{y}.$

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28. If $x=1$, $y=-1$ and $z=-1$, then the value of the expression $\frac{x+y+z}{x^2+y^2+z^2}$ will be

A. $-\frac{1}{3}$
B. $\frac{1}{3}$
C. 0
D. 1

29. If $x+y=7$ and $x-y=3$, then the value of x^2-y^2 will be equal to

A. 40
B. 21
C. 10
D. 4

30. The sum of surds given as $\sqrt{x+1}+\sqrt{4(x+1)}$ will be equal to

A. $5\sqrt{x}+5$
B. $5\sqrt{x+1}$
C. $3\sqrt{x}+3$
D. $3\sqrt{x+1}$

31. The simplest form of $\frac{2}{2-\sqrt{3}}$ will be

A. $\frac{1}{-\sqrt{3}}$
B. $2\sqrt{3}$
C. $4-2\sqrt{3}$
D. $4+2\sqrt{3}$

32. The expression $(a-1)x+(a-1)$ is equal to

A. $(a-1)x$.
B. $(a-1)(x+1)$.
C. $(a-1)^2(x-1)$.
D. $(a-1)^2(x+1)$.

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33. One of the factors of $(p+r)^2x^2-1$ will be

A. $(p+r)-x$
B. $(p+r)-1$
C. $(p+r)+1$
D. $(p+r)x+1$

34. The factors of $x^2-x+\frac{1}{4}$ are

A. $x-\frac{1}{2}$ and $x+\frac{1}{2}$
B. $x-\frac{1}{4}$ and $x+\frac{1}{4}$
C. $x-\frac{1}{2}$ and $x-\frac{1}{2}$
D. $x-\frac{1}{4}$ and $x-\frac{1}{4}$

35. If the expression x^3-x^2-p is divided by $x+1$, then the remainder is 1. The value of p will be

A. -3
B. -2
C. 2
D. 3

36. If $p(x)=(x-1)(x+2)(x-3)$, then one of the zeros of the $p(x)$ will be

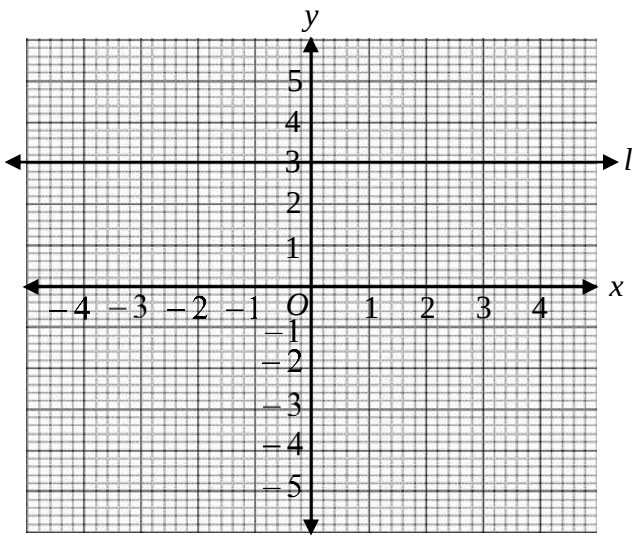
A. -3
B. -1
C. 2
D. 3

37. In which quadrant do both the x-coordinate and y-coordinate have positive signs?

A. First quadrant
B. Second quadrant
C. Third quadrant
D. Fourth quadrant

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38. In the given graph, the equation of the line l will be



A. $y = -3$
B. $x = -3$
C. $y = 3$
D. $x = 3$

39. Which of the following equations satisfy the point $(2, 3)$?

A. $y = 3x$
B. $y = x + 2$
C. $y = 2x + 1$
D. $y = x + 1$

40. An example of a diagonal matrix is

A. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.
B. $\begin{bmatrix} 2 & 2 \\ 4 & 4 \end{bmatrix}$.
C. $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$.
D. $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$.

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41. If $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, then the matrix $-2A$ will be equal to

A. $\begin{bmatrix} -4 & -2 \\ -2 & -4 \end{bmatrix}$.

B. $\begin{bmatrix} -4 & 0 \\ 0 & -4 \end{bmatrix}$.

C. $\begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$.

D. $\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$.

42. The sum of the matrices $\begin{bmatrix} -1 & 2 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}$ will be

A. $\begin{bmatrix} 2 & 4 \\ 0 & 0 \end{bmatrix}$.

B. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

C. $\begin{bmatrix} -3 & 3 \\ 0 & 0 \end{bmatrix}$.

D. $\begin{bmatrix} -2 & -4 \\ 0 & 0 \end{bmatrix}$.

43. If the determinant of the matrix $\begin{bmatrix} a & 5 \\ 0 & 1 \end{bmatrix}$ is 4, then the value of a will be

A. 1
B. 4
C. 5
D. 6

44. The determinant of a matrix is represented as $\begin{vmatrix} 2 & 2 \\ 1 & 1 \end{vmatrix}$.

Based on the given determinant, the matrix would be

A. symmetric.
B. diagonal.
C. singular.
D. null.

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45. Which of the given matrix can have a multiplicative inverse?

A. $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

B. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

C. $\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$

D. $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$

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END OF PAPER

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